Projection Obstruction Theory: Retraction Nonlinearity, ℓ^1 Rigidity, and Density Scaling

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Abstract

Projection-based approximations — such as mean-field theory, product-state ansatze, and factorized variational methods — replace correlated states with product states via a nonlinear retraction. This induces a structural discrepancy between the true dynamics and the projected dynamics.

We analyze this discrepancy in a Banach presheaf framework and establish three results. **(1)** For a C^2 retraction Π onto a submanifold M, the induced dynamics $S_t = \Pi \circ T_t|_M$ is, under a non-degeneracy condition on the Jacobian of the projected generator, not conjugate to any linear semigroup (Theorem 3.1). **(2)** Under strict locality, additivity, and symmetry axioms, the ℓ^1 norm of the presheaf coboundary is the canonical additive diagnostic for this discrepancy, characterized as the initial object in an appropriate category of seminorms (Theorem 4.2). **(3)** For finite-range, finite-dimensional lattice systems with non-commuting interactions, the induced defect exhibits non-vanishing density over a finite time interval independent of system size, as constrained by Lieb–Robinson bounds (Theorem 5.1).

We further observe that Axioms 1–4 are not arbitrary: they characterize the minimal structural conditions under which a local agent can consistently detect, aggregate, and act upon defects (Section 1.2). This grounds the ℓ^1 diagnostic in an operational criterion — the existence of consistent local observers — rather than in unmotivated mathematical assumptions.

These results formalize the geometric, categorical, and dynamical structure of projection-induced errors in factorized many-body approximations. The framework is finite-dimensional and finite-time; no claims are made regarding asymptotic or thermodynamic limits.

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1. Introduction

Projection-based approximations are central to many-body physics and computation. Methods such as mean-field theory, Hartree–Fock, and product-state variational ansatze replace a correlated state with a product of local states, thereby reducing an exponentially large description to a tractable one. This replacement is implemented by a nonlinear map — formally, a retraction — onto a low-dimensional manifold of factorized states. While the practical success of these methods is well established, the structural effect of this retraction on

the underlying dynamics is less clearly understood: what information is systematically discarded, how it propagates, and how it should be measured in a representation-independent way.

In this work, we isolate and analyze the discrepancy introduced by projection as a geometric and combinatorial object. We show that composing a linear many-body evolution with a nonlinear retraction produces dynamics that — under a non-degeneracy condition — cannot be reduced to any linear flow, and that the resulting local inconsistencies assemble into a presheaf coboundary with a canonical additive structure. Under natural locality and symmetry conditions, this leads to a distinguished ℓ^1 diagnostic for projection-induced defects, and for finite-range systems we prove that this defect persists with non-vanishing density over a finite time interval independent of system size.

The goal is not to propose a new physical theory, but to provide a precise structural account of the error introduced by factorization.

We prove three structural theorems: 1. **Geometric Obstruction (Theorem 3.1):** The projected dynamics is not C^2-conjugate to any linear semigroup when the projected generator has non-constant Jacobian. 2. **Additive Universality (Theorem 4.2):** Subject to strict additivity and symmetry axioms, the ℓ^1 coboundary norm is the uniquely determined diagnostic, up to a global scalar. Relaxing these axioms admits alternative norms (e.g. ℓ^2). 3. **Density Persistence (Theorem 5.1):** The per-bond obstruction density is $\Omega(1)$ in system size over a time window $O(1/J)$, with constants depending only on local dimension, interaction strength, and active bond fraction.

Scope. The framework applies rigorously to Banach presheaves over finite posets. We avoid speculative physics or infinite-time limits; the density scaling bound is finite-time only.

Conventions. Norms are denoted $\|\cdot\|$ with appropriate subscripts; scalar absolute values are denoted $|\cdot|$.

1.1 Relation to Companion Work

The additive diagnostic used throughout this paper — the ℓ^1 coboundary norm — is not introduced ad hoc. It is the terminal output of a two-stage classification program establishing ℓ^1 rigidity at progressively richer structural levels:

- **Stage 1 (combinatorial):** In "Coordinate-Wise Additivity and the ℓ^1 Norm on Finite Graph Cochains" [8], local additivity over disjoint coordinate supports and edge separability on finite graph cochains already force an ℓ^1 geometry. No Banach space structure or dynamics are required.
- **Stage 2 (functional-analytic):** In "The Free ℓ^1 Seminorm on Banach Presheaf Coboundaries" [6], the same rigidity persists when the cochain spaces are replaced by Banach presheaves and coordinate additivity is replaced by structural compatibility (functoriality under bounded linear maps).

By the classification results of these preceding papers, any diagnostic satisfying locality, local additivity, and compatibility with the underlying linear structure must be proportional to the ℓ^1 coboundary norm. Accordingly, the defect quantity analyzed here is not chosen, but structurally determined by these constraints.

The present paper (Stage 3, dynamical) builds on that classification by showing that projection-induced dynamics naturally generate nontrivial coboundary defects, and that these defects persist with non-vanishing density under finite-range evolution. Together, the three papers establish that additive, structure-preserving measurements of local defects are uniquely forced to be ℓ^1 across combinatorial, functional, and dynamical settings, and that projection dynamics generically produce nonzero instances of such defects.

1.2 Observer-Theoretic Justification of Axioms

Axioms 1–4, which determine the ℓ^1 coboundary norm (Theorem 4.2), are not arbitrary choices. They arise as the minimal structural conditions required for the existence of consistent local observers — agents that detect, aggregate, and act upon local inconsistencies in a many-body system.

Axiom 1 (Coboundary Locality) follows from the requirement that an observer can detect a defect by inspecting a single edge without access to the global state. Any diagnostic that requires global information to evaluate a single inconsistency is operationally unavailable to a local agent.

Axiom 2 (Faithfulness) follows from the requirement that the diagnostic is nontrivial: an observer that reports zero defect when genuine inconsistencies exist is not an observer in any useful sense.

Axiom 3 (Contractive Monotonicity) follows from the requirement that local corrections cannot increase the total measured defect. If an observer applies a local repair (a contractive map on one fiber), the resulting global defect must not exceed the original. Violation of this condition would mean that fixing a local inconsistency creates worse global inconsistency — making the diagnostic adversarial to its own use.

Axiom 4 (Symmetry and Coproduct Compatibility) follows from the requirement that observers cannot distinguish structurally isomorphic subsystems, and that the diagnostic on a composite system with no inter-component edges is the sum of diagnostics on the components. This is the additivity condition: defects on disconnected regions aggregate without cross-terms or cancellation.

Taken together, these conditions characterize what it means for a defect diagnostic to be *usable by a local agent*: locally evaluable, nontrivial, stable under local repair, and compositional over independent subsystems. By Theorem 4.2 (and the classification results of the companion papers [8, 6]), the ℓ^1 coboundary norm is the unique diagnostic satisfying these conditions.

This reframing does not weaken the conditionality of Theorem 4.2: the conclusion remains that ℓ^1 is forced *given* these axioms. What it does is ground the axioms themselves in an operational criterion — the existence of consistent local observers — rather than presenting them as unmotivated mathematical assumptions. Relaxing the axioms (e.g. permitting cancellation between defect channels) still admits alternative norms such as ℓ^2, but at the cost of the observer-consistency interpretation.

1.3 Observer-Theoretic Derivation — Expanded

The four axioms of the coboundary norm (Locality, Faithfulness, Contractive Monotonicity, Coproduct Compatibility) are not arbitrary mathematical assumptions. They are the minimal structural conditions required for the existence of consistent local observers.

Consider an observer $\mathcal{O}$ defined as a subsystem $U \subset G$ whose internal section $s|_U$ must remain stable under the global dynamics T. For $\mathcal{O}$ to qualify as a persistent observer, four requirements are necessary:

1. **Local detection** (forces Axiom 1: Coboundary Locality). The observer must detect contradictions using only local information — comparing values on adjacent vertices. A nonlocal norm (e.g., ℓ^∞, which only monitors the single worst edge) fails to detect distributed errors that accumulate and destroy the observer section.

2. **Additive aggregation** (forces Axiom 4: Coproduct Compatibility). The observer must aggregate defects additively across its boundary. A norm permitting cancellation (e.g., ℓ^2, where the Euclidean norm does not aggregate additively over independent components) creates a "quiet period" where accumulated errors are invisible, followed by sudden catastrophic section fracture.

3. **No hidden cancellation** (forces Axiom 2: Faithfulness). Every nonzero edge defect must contribute positively to the total defect measure. If any edge can report zero defect when its actual defect is nonzero, the observer's corrective mechanism has a blind spot.

4. **Non-amplifying repair** (forces Axiom 3: Contractive Monotonicity). Local corrections applied by the observer must not increase the global defect. If repairs can amplify errors, a feedback cascade destabilizes the observer.

These four conditions are independently necessary: relaxing any single axiom produces a distinct failure mode for observer stability (see the axiom relaxation analysis in the companion work (forthcoming)).

1.4 Conceptual Overview

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Full correlated state space X
        |
        |  C^2 retraction Pi
        v
Product-state manifold M
        |
        |  induced curvature term D^2Pi(Am, .)
        v
Projected dynamics S_t = Pi . T_t|_M
        |
        |  local incompatibilities across regions
        v
Coboundary defects delta_e(s)
        |
        |  additive aggregation (Axioms 1-4)
        v
Total obstruction  I(s) = Sum ||delta_e(s)||
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2. Structural Framework

We work with Banach-valued presheaves over finite posets, following standard categorical conventions [2].

2.1 Presheaves over Finite Posets

Let $(P, \leq)$ be a finite poset with covering relations $E_{\text{cov}}(P) = \{(a, b) : a < b,\ \nexists c \text{ with } a < c < b\}$. A **Banach presheaf** F over P assigns to each $x \in P$ a Banach space $F(x)$ and to each $(a, b) \in E_{\text{cov}}$ a contractive linear restriction map $r_{b,a} : F(b) \to F(a)$.

2.2 The Coboundary

For each edge $e = (a, b)$, the **coboundary** is $\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a)$. A section descends if $\delta_e(s) = 0$ for all e. The ℓ^1 coboundary norm is $I(s) = \sum_{e \in E_{\text{cov}}} \|\delta_e(s)\|$.

2.3 Linear Semigroups and Projected Dynamics

A C_0 **contraction semigroup** on a Banach space X is $\{T_t\}_{t \geq 0}$ with generator $(A, \mathcal{D}(A))$. A C^2 **retraction** is a C^2 idempotent $\Pi : X \to M \subset X$ with M a regular C^2 submanifold. If $\{T_t\}$ does not preserve M, we define the **projected semigroup** $S_t = \Pi \circ T_t|_M$ mapping $M \to M$ locally.

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3. Theorem I: Nonlinear Obstruction to Linear Conjugacy

3.1 Statement

Theorem 3.1 (Projection Non-Dilability). Let X be a Banach space, $\{T_t\}$ a C_0 contraction semigroup with generator $(A, \mathcal{D}(A))$, and $\Pi : X \to M \subset X$ a C^2 retraction onto a C^2 submanifold $M \subset \mathcal{D}(A)$. Suppose:

(a) *(Non-invariance)* There exists $t > 0$ such that $T_t(M) \not\subset M$.

(b) *(Non-degeneracy)* The induced vector field $B(m) = D\Pi(m)Am$ has non-constant Jacobian $DB(m)$ in any affine coordinate chart compatible with a flat connection.

Then there does not exist a Banach space Y, a C^2 embedding $\iota : M \hookrightarrow Y$, and a C_0 contraction semigroup $\{\tilde{T}_t\}$ on Y such that $S_t = \iota^{-1} \circ \tilde{T}_t \circ \iota$.

3.2 Preliminary Lemma

Lemma 3.2 (Parallel fields on flat connections). Let (M, ∇) be a connected Banach manifold equipped with a flat affine connection ($R^\nabla = 0$). If a $(1,1)$-tensor field S on M satisfies $\nabla S = 0$, then S is constant in any affine coordinate chart.

Proof. Since $R^\nabla = 0$, the exponential map provides local affine coordinates where the Christoffel symbols vanish. In these coordinates, $\nabla S = 0$ reduces to $\partial_i S^j_k = 0$, making components constant. Affine transitions preserve this constancy globally. □

3.3 Proof of Theorem 3.1

Step 1: Linear flows are affinely flat. A linear semigroup $\{\tilde{T}_t\}$ with generator $\tilde{A}$ satisfies $\tilde{A}(y) = \tilde{A}y$. The canonical flat connection $\tilde{\nabla}$ on Y has $R^{\tilde{\nabla}} = 0$, and $\tilde{\nabla}^2 \tilde{A} = 0$.

Step 2: Conjugacy transfers flatness. Suppose $\iota : M \hookrightarrow Y$ is a C^2 embedding conjugating S_t to $\tilde{T}_t$. The pullback connection $\nabla = \iota^* \tilde{\nabla}$ is flat on M (since ι is C^2, the pullback preserves the vanishing of curvature), and $\nabla^2 B = 0$ for the generator $B = \iota_*^{-1} \tilde{A}$ of S_t.

Step 3: The projected generator has non-constant derivative. The projected generator is $B(m) = D\Pi(m)Am$. Differentiating yields:

$$\nabla_v B = D^2\Pi(m)(Am, v) + D\Pi(m)Av$$

If $\nabla^2 B = 0$, then $\nabla_v B$ would be a parallel tensor field on (M, ∇), and by Lemma 3.2, it would be constant in any affine coordinate chart. But Hypothesis (b) states that $DB(m)$ is non-constant in every such chart. This is a contradiction. □

4. Theorem II: Additive Universality of the ℓ^1 Diagnostic

Theorem 3.1 demonstrates that geometric nonlinearity is inevitably produced by the retraction. The next question is how to measure the resulting defect. Theorem 4.2 establishes that, given strict additivity and symmetry constraints, the ℓ^1 norm is the uniquely forced diagnostic. The conclusion is conditional on these axioms; relaxing them admits other norms.

4.1 Diagnostic Systems and the Category of Seminorms

Let $C^1(P, F) = \bigoplus_{e \in E_{\text{cov}}} F(a_e)$ be the space of edge defects. We consider seminorms $J : C^1(P, F) \to \mathbb{R}_{\geq 0}$ satisfying:

Axiom 1 (Coboundary Locality). $J_P(s) = \sum_e w_e \|\delta_e s\|$, with local weights $w_e > 0$.

Axiom 2 (Faithfulness). $J_P(s) = 0 \iff s$ descends.

Axiom 3 (Contractive Monotonicity). For local contractions, the norm weakly decreases.

Axiom 4 (Symmetry and Coproduct Compatibility). Edge weights are uniform by edge-exchange symmetry, and the norm splits linearly over disjoint graph unions.

4.2 Statement

> **Theorem 4.2 (ℓ^1 Additive Universality).** The ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is the initial object in the category of faithful seminorms on $C^1(P, F)$ satisfying Axioms 1–4, in the sense that any other such seminorm factors uniquely through I up to a positive scalar. Consequently, any diagnostic satisfying these axioms is proportional to I:
>
> $$J(s) = c \sum_e \|\delta_e(s)\|$$
>
> for some constant $c > 0$.

Proof Sketch. Any diagnostic J_P satisfying the axioms must assign constant weights to local defects due to edge-exchange isomorphism (Axiom 4). Concretely, edge-exchange symmetry implies that all edge weights w_e are equal, since any permutation of edges that preserves graph structure must leave J invariant. The Kan extension $\text{Lan}_\pi d(s)$ realizes the universal space of local defects. Additivity over disjoint unions (Axiom 4) forces the seminorm to decompose as a sum over edges, eliminating ℓ^2 or ℓ^∞ alternatives which fail additivity over coproducts. Thus ℓ^1 is uniquely selected. □

Remark. The ℓ^1 norm is uniquely selected **given strict additivity and symmetry constraints**, which may or may not be physically appropriate depending on context. The real content of this theorem is structural: ℓ^1 corresponds to additive aggregation of independent defects. In physical contexts where defect channels are not independent (e.g. when correlations between edges matter), alternative diagnostics such as Hilbert–Schmidt are natural.

Observer-theoretic interpretation. As discussed in Section 1.2, Axioms 1–4 have a natural operational reading: they are the conditions under which a *local observer* can consistently detect and aggregate defects. Under this reading, Theorem 4.2 states that any defect diagnostic usable by a consistent local observer must be proportional to the ℓ^1 coboundary norm. The conditional structure is preserved: the theorem does not claim ℓ^1 is universally forced, but that it is forced for any system admitting consistent local observation of defects.

By the classification results of the preceding papers [8, 6] — first at the combinatorial level (local additivity over disjoint supports) and then at the Banach level (functoriality under bounded linear maps) — any diagnostic satisfying locality, additivity, and structural compatibility must be proportional to the ℓ^1 coboundary norm. Accordingly, the defect quantity analyzed here is not chosen, but structurally determined by these constraints.

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5. Theorem III: Finite-Time Density Persistence

5.1 Setup

Consider N sites on a finite graph Λ with local Hilbert spaces ($\dim \mathcal{H}_x \leq d < \infty$), bounded Hamiltonian $H = \sum_e H_e$ with $\|H_e\|_{\mathrm{op}} \leq J$, and retraction $\Pi(\rho) = \bigotimes_x \mathrm{Tr}_{\bar{x}}(\rho)$. Local Hamiltonians satisfying the non-commutativity requirement below are generic in the space of two-local interactions [3].

Define the commutator:

$$[H_e, \rho_a \otimes \rho_b] = H_e(\rho_a \otimes \rho_b) - (\rho_a \otimes \rho_b)H_e$$

5.2 Explicit Rate Bound

Lemma 5.2 (Local obstruction rate). For a single bond $e = (a, b)$ with $[H_e, \rho_a \otimes \rho_b] \neq 0$, let $\gamma_e = \|[H_e, \rho_a \otimes \rho_b]\|_1/d^2$. Then $\gamma_e > 0$. For t small:

$$C_e(t) \geq \gamma_e t - O(J^2 t^2).$$

Proof. On finite-dimensional spaces, the map $t \mapsto \rho(t) - \Pi(\rho(t))$ is analytic. Expanding to first order, the leading term is $-it[H_e, \rho_a \otimes \rho_b]$, whose trace norm is $\gamma_e d^2/d^2 = \gamma_e$. The remainder is bounded by $O(J^2 t^2)$ via the operator norm bound $\|H_e\|_{\mathrm{op}} \leq J$. Since $\gamma_e > 0$ by hypothesis, the linear term dominates for $t < \gamma_e/(KJ^2)$ where K is a computable constant depending only on d. $\square$

5.3 The Density Bound

Theorem 5.1 (Finite-Time Density Persistence). Let Λ be a finite graph with N sites and finite-dimensional local Hilbert spaces ($\dim \leq d$). Let $H = \sum_e H_e$ be a bounded finite-range Hamiltonian with $\|H_e\|_{\mathrm{op}} \leq J$, and suppose a fraction $\eta > 0$ of edges satisfy $[H_e, \rho_a \otimes \rho_b] \neq 0$ for an initial product state $\rho(0) = \bigotimes_x \rho_x$.

Then there exist constants $c > 0$ and $t_0 > 0$, depending only on local dimension d, interaction strength J, and active fraction η (but not on system size N), such that

$$\frac{C(t)}{N} \geq c \quad \text{for all } t \in [t_1, t_2] \subset (0, t_0]$$

where $t_0 = O(1/J)$.

Proof sketch. By Lemma 5.2, each active bond e generates a trace-norm defect $C_e(t) \geq \gamma_e t - O(J^2 t^2)$ for small t. By the Lieb–Robinson bound [1, 4, 5], for times $t < R/v_{\mathrm{LR}}$ (where R is the interaction range and v_{LR} the Lieb–Robinson velocity), the reduced density matrix on any bond is approximated by its isolated evolution up to an error exponentially small in $(v_{\mathrm{LR}} t - R)$. Concretely, the trace-norm error from truncating the causal cone satisfies $\|\rho_e^{\mathrm{exact}}(t) - \rho_e^{\mathrm{local}}(t)\|_1 \leq C_{\mathrm{LR}} e^{-\mu(R - v_{\mathrm{LR}} t)}$ for constants $C_{\mathrm{LR}}, \mu > 0$ depending on the lattice geometry.

Within this causal window, each bond's defect $C_e(t)$ is controlled by the local evolution and cannot be canceled by correlations from distant bonds, because such correlations have not yet propagated to the bond. Summing over the ηN active bonds and noting that $\gamma_e \geq \gamma_{\min} > 0$ (where $\gamma_{\min}$ depends on J and d), we obtain $C(t) \geq \eta N(\gamma_{\min} t - K J^2 t^2)$ for $t \in (0, \gamma_{\min}/(2KJ^2))$. Dividing by N gives $C(t)/N \geq \eta \gamma_{\min} t/2 > 0$ on a non-trivial interval, with all constants independent of N. □

Remark on the time window. The interval $[t_1, t_2]$ scales as $O(1/J)$: the defect appears immediately (linear growth at rate γ_e), dominates the quadratic correction for $t \ll 1/J$, and the Lieb–Robinson causal cone remains localized for $t \ll R/v_{\mathrm{LR}} \sim 1/J$. Beyond this window, many-body correlations propagate across bonds and the argument does not apply.

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6. Worked Examples

6.1 Example 1: Two Qubits (Entangling)

Consider two qubits with $H = J(X \otimes X)$ and initial state $|00\rangle$. The commutator $[H, |00\rangle\langle 00|]$ is strictly non-zero. The full evolution yields:

$$|\psi(t)\rangle = \cos(Jt)|00\rangle - i \sin(Jt)|11\rangle$$

$$\rho(t) = \cos^2(Jt)|00\rangle\langle 00| + \sin^2(Jt)|11\rangle\langle 11| + (\text{off-diagonal terms})$$

The product retraction isolates marginals:

$$\rho_A(t) = \rho_B(t) = \cos^2(Jt)|0\rangle\langle 0| + \sin^2(Jt)|1\rangle\langle 1|$$

$$(\Pi\rho)(t) = \cos^4(Jt)|00\rangle\langle 00| + \sin^4(Jt)|11\rangle\langle 11| + \cos^2(Jt)\sin^2(Jt)(|01\rangle\langle 01| + |10\rangle\langle 10|)$$

The trace norm defect $C(t) = \|\rho - \Pi\rho\|_1 > 0$ for $t > 0$. This witnesses non-product structure in $\rho(t)$: the true state cannot be written as a tensor product of marginals, and the defect quantifies this failure.

6.2 Example 2: Transverse-Field Ising Model

The 1D Ising model $H = -J\sum_i Z_i Z_{i+1} - h\sum_i X_i$ with $|+\rangle^{\otimes N}$ produces strictly positive initial rates $C_e(t) = \gamma_e t + O(J^2t^2) > 0$, consistent with the linear-in-t scaling established by Lemma 5.2. By translational invariance, per-bond density is insensitive to N for $t \ll N/v_{\mathrm{LR}}$.

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7. Discussion

This work isolates a structural feature common to projection-based approximations: the act of restricting a linear many-body evolution to a nonlinear manifold introduces an intrinsic obstruction that is both geometrically unavoidable and locally detectable. The three results established here — nonlinear non-dilability of the projected flow, additive universality of the ℓ^1 coboundary diagnostic under symmetry and locality constraints, and finite-time persistence of defect density in finite-range systems — together provide a coherent description of how information is lost under factorization and how that loss propagates. Importantly, these statements are finite-dimensional and finite-time, and rely only on structural properties of the retraction and locality of interactions, rather than on model-specific features. In this sense, the framework does not compete with existing approximation methods, but rather clarifies the limits they necessarily obey.

Looking ahead, just as differential forms decompose into local and global topological pieces, resolving $C^1(P, F)$ defects suggests a generalized **Hodge decomposition** for dynamical lattice models:

$$\delta s = d\alpha + \delta\beta + h$$

where $d\alpha$ isolates removable trivial projection errors, $\delta\beta$ captures dynamical correlation flux, and h localizes persistent obstruction invariants (harmonic forms). This decomposition, together with its duality structure, is developed in companion papers [6, 7].

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Appendix A: Derivation Chain — From Observers to the Born Rule

This appendix outlines a broader research program and is not required for the results of this paper.

The framework developed across Papers 000–002 and the companion work on discrete renormalization admits a single connected derivation chain linking observer consistency to the Born rule of quantum mechanics. We state this chain explicitly to make the logical structure transparent.

A.1 The Three-Layer Structure

Layer 0 (Observer Existence -> Axioms). Any discrete dynamical system admitting stable, self-consistent observers must satisfy: - Coboundary Locality (Axiom 1): defect detection must be local. - Faithfulness (Axiom 2): the diagnostic must be nontrivial. - Contractive Monotonicity (Axiom 3): local repairs cannot amplify global error. - Coproduct Compatibility (Axiom 4): defects on disjoint regions aggregate without cancellation.

These are derived in Sections 1.2 and 1.3 of this paper from operational requirements on local agents.

Layer 1 (Axioms -> ℓ^1 Uniqueness). Under Axioms 1–4, the ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is the unique faithful seminorm up to positive scalar (Theorem 4.2 of this paper; supported by the classification results of [6, 8]).

Layer 2 (ℓ^1 Dynamics -> Emergent Quantum Mechanics). ℓ^1 dynamics on discrete constrained graphs produce (forthcoming): 1. Non-factorizable bounded convolutions (global ℓ^1 constraints prevent classical real-space decomposition). 2. Fourier factorization: locality forces diagonalization into the eigenbasis of the transport operator. 3. Complex amplitudes: causal asymmetry and directed transport force the eigenstates into the complex domain, producing $U(1)$ phase. 4. Haar measure: observer gauge ignorance (inability to access the universal phase origin) forces integration over the unique shift-invariant measure on $U(1)$. 5. Born rule: Parseval's theorem under the Haar measure yields $P = |\psi|^2$. 6. IR Hilbert space: RG block-averaging forces the macroscopic transport operator toward normality (a circulant matrix), producing the Hilbert space as the emergent IR representation.

A.2 The Formal Chain

$$\text{Observers exist} \xrightarrow{\S 1.2} \text{Axioms 1–4} \xrightarrow{\text{Thm 4.2}} \ell^1 \xrightarrow{\text{(forthcoming)}} \text{Non-factorization} \xrightarrow{\text{DFT}} U(1) \xrightarrow{\text{Haar}} P = |\psi|^2 \xrightarrow{\text{RG}} \mathcal{H}$$

Each arrow represents a mathematical implication. The chain is falsifiable at every joint: if any link fails under computational or mathematical scrutiny, the downstream structure collapses.

A.3 Scope and Caveats

This derivation chain is stated at the level of a research program, not a completed proof. Layers 0 and 1 rest on the rigorous results of this paper and its companions. Layer 2 contains both established results (Fourier factorization, Parseval invariance) and conjectural steps (the universality of the RG attractor, the parameter-free emergence of α). The chain is presented to make the logical dependencies explicit and to identify where further formal work is needed.

The cross-reference structure is as follows. The Layer 0 to Layer 1 transition is established in Section 1.3 of this paper and in the axiom relaxation analysis of the companion work (forthcoming). The Layer 1 to Layer 2 transitions — non-factorization, Born rule emergence, and the RG attractor to Hilbert space — are developed in forthcoming work on discrete renormalization and the axiomatic gap.

The conditionality of all claims is preserved: ℓ^1 is forced *given* the axioms; the axioms are motivated *given* observer consistency; the Born rule emerges *given* ℓ^1 dynamics on frustrated graphs. No claim of unconditional physical necessity is made.